

1. Rule Discovery

Heterogeneous Policy Docs



Drug Directory
Health Insurance
Payment Policy



Medical Service
Price List
.....

Table-Aware Discovery



Table-Aware Understanding
Implicit Rule Discovery
IRR Construction

IRR

Rule Category

Restricted Medication Use

Constraint Object

Selumetinib Capsules

Constraint Condition

only for patients
aged 3 years or older with
neurofibromatosis type 1 (NF1),
symptomatic, inoperable
plexiform neurofibromas.

2. Agentic Rule Compilation (Core)

Step 2.1 Terminology Normalization + Similar Rule Retrieval

Step 2.2 Semantic Decomposition

Input: IRR (Constraint Condition)

Output: Atomic Constraints + Logical Structure

Constraint Condition

only for patients
aged 3 years or older with
neurofibromatosis type 1 (NF1),
symptomatic, inoperable
plexiform neurofibromas.

Atomic Constraints

- C1 Age $\geq$ 3 years
- C2 Diagnosis = NF1 (Neurofibromatosis type 1)
- C3 Symptomatic Plexiform Neurofibroma
- C4 Inoperable Plexiform Neurofibroma

Logical Structure (L)

$L = C1 \wedge C2 \wedge C3 \wedge C4$
(All conditions must be met)

Step 2.3 Tool-grounded Code Generation

Input:

- IRR & Atomic Constraints & Logical Structure (from 2.2)
- Retrieved Functions & Similar Rules (Context)

IRR

Atomic
Constraints

Logical
Structure

Function Library

Data Access Layer

```
get_val(patient, field)    # get a value from EMR  
filter_list(table, "conds") # filter records  
.....
```

Rule Operator Layer

```
is_compare(a, op, b)       # e.g.,  $\geq$ ,  $=$ ,  $\neq$   
list_contains(lst, x)      # membership check  
.....
```

Similar Rule Examples (Retrieved)

- Rule: Age $\geq$ 18 and Diagnosis = Diabetes
Code: `get_val(...); is_compare(...); list_contains(...)`
- Rule: Diagnosis = Hypertension and Symptomatic
Code: `list_contains(...); get_val(...); ...`

LLM



- Select functions
- Fill parameters
- Compose logic



output:
**Executable
Audit
Program
(Python)**

3. Rule Grounding

Rule execution on EMRs

Python Audit Program



Execution on EMR

Patient EMR(JSON)

Audit Result (Example)

Violation Status: **PASS**

Violation Reason:

Executable Evidence:

```
{  
  "age_years": 45,  
  "diagnosis_codes": ["NF1"],  
  "pn_symptomatic": true,  
  "pn_inoperable": true  
}
```

Inspection Result Alignment



Inspection
Records

Column
Mapping &
Normalization

Rule-Instance
Matching
(by patient id
& rule)



Consistency
Analysis



Execution
results